Published-IC reference arm: first pilot cell (pre-freeze), 2026-10-30

Card `XP-20261024-published-ic-reference-arm`. **Status: pre-freeze pilot.** `PROTOCOL.md` is still a DRAFT and says no cell is timed before freeze, so **nothing here is a Register-E row** and nothing here supports a "vs published index calculus" claim. Labels: *measured* (a recorded value), *derived* (arithmetic shown), *projection* (derived from fewer measurements than the cell it names).

1. Answer

A published-method cell ran. On Koblitz K_0 n = 9 (r = 127, m = 3, standard x-base of dimension 4), whole solve, 8 public targets, 8/8 verified `[d]G = Q`, the msolve arm costs about 10^12 instructions retired per target (measured, A/A within 0.08 %), against a lab arm too small to resolve at the instruction level (at most 3.75 x 10^6 per target, the harness noise floor; ratio at least 2.8 x 10^5). Priced in GAE (derived, 605 instructions per GAE) the arm is at S ~ 1.5 x 10^8 against rho 5.9 and the lab arm's 36.1 (solver uncharged). The order (msolve arm above the lab arm) holds across the AGENTS.md section 5a pricing band. The n = 13 whole solve did **not** run: one system takes 5.70 x 10^12 instructions (544 s wall under load), and the lab arm's recorded K13 stream needs about 36 of them per target.

This says what *untuned, generic msolve F4 on the unspecialised Weil-descended system* costs. It does not say what the best published pipeline costs: no hybrid (variable specialisation) sweep, no l sweep, no T-invariants, no Galbraith-Gebregiyorgis subspace, no WDSat were run (section 5). The lab's own inherited-F4 solver splits on variables and is itself close to a published hybrid approach, so a large ratio here is as much a statement about msolve's default on these systems as about the lab. The distance to the *tuned* published arm is unmeasured.

2. Arm as run (and where it deviates from `PROTOCOL.md`)

- Instance: `icv1-f2m9-t5-81e744be` (Koblitz K_0, `KoblitzCurve::new(0, 9)`, #E = 508, r = 127, cofactor 4), base `binary-standard-subspace:dimension=4` (x in span(1, z, z^2, z^3)), m = 3; targets: the 8 public targets of ecbench's `target_seed 202610111901` law. Lab arm, rho arm and published arm are arms of **the same session** `ECBS1h3c8ebd974409` (same instance, same targets, same query stream), `ecbench` built from this tree.
- Published arm = `pdp3-binary:m=3,engine=msolve-s4` (new, `src/bin/ecbench/msolve_s4.rs`): the `e_i`-symmetrised Weil-descended Semaev `S_4` of `binary_semaev_s4` (descended `S_4`, quadratic in `e_1,e_2,e_3`, plus the cubic correspondence `e_i = sigma_i(x)`), 3l + (6l-3) = 33 unknowns at l = 4, field equations appended, solved by `/opt/homebrew/bin/msolve` 0.9.5, `-g 2 -t 1`, one process per system. Roots from the reduced basis (section 4), lifted through the framework's `lift_abscissae`, relation sign-resolved and re-added in the group, then the pipeline's rank, linear algebra and `[d]G = Q` verification. A second, tuning-probe form `MSOLVE_S4_FORM=x` substitutes `e_i = sigma_i(x)` and solves over the 3l x-bits only; it ran in the probe, not in a session.
- Deviations from `PROTOCOL.md`, all recorded rather than silent:
  1. `S_3`-symmetrisation only; **no 2-torsion `T` invariants** (`binary_semaev_s4.rs` has none; PROTOCOL section 2′s "T-symmetrised" and section 1′s "`e_i`-symmetrised" disagree, see defect D6).

2. msolve **0.9.5** (sha256 `eb640c7a472d894a...`), not the pinned 0.6.5; no version A/A.
3. **No tuning sweep** (`l`, `LP`, `K`, specialisation); **no Galbraith-Gebregiyorgis subspace**; **no WDSat**.
4. Pricing is **not** the protocol's Callgrind `Ir`: no Valgrind on this host. Instructions retired from `/usr/bin/time -l` (Apple M4 Pro, aarch64) stand in; the protocol's cross-unit control was **not** run (section 3).
5. Isolation: macOS, `ecbench --cpus none`, every record is L0 (PROTOCOL asks L1). Counted metrics are unaffected; host load was 10-50, so wall times in the records are not claims and are not used.

3. Pricing, and the cross-unit control that was not met

- msolve instructions retired: `/usr/bin/time -l` around each msolve call (needs the sandbox off for a sysctl). Recorded per call in the session's `solver.extra.msolve_ir`. Run to run on the same 8 targets: 8 of 8 targets within 0.08 % (`runs/C9` against `runs/D9`), total 8.5403 x 10^12 against 8.5366 x 10^12 (*measured*).
- Instructions per GAE (*derived*): two rho-negation sessions on K_0 n = 37, 1 target and 5 targets (`runs/CAL1`, `CAL5`; CAL1 repeated as `CAL1b`: 1.5 x 10^6 apart). Parent instructions retired 2 803 163 935 - 2 775 941 181 = 2.722 x 10^7 for 55 046 - 10 022 = 45 024 GAE: **604.6 instructions per GAE**. It contains the walk pipeline's own overhead, and was measured at n = 37, where an addition costs more than at n = 9; both push the ratio up, so msolve's GAE count here is an *under*-statement. This is **not** the protocol's control (a native-counted lab solver priced two ways); it replaces it for this pilot only, and carries no 1.5x tolerance claim.
- The lab arm's solver is counted in word XORs, which `ecbench` leaves **unpriced** (S is a lower bound, as the leaderboard says), and `ecbench` has no pinned word-XOR ratio for n = 9 or 13. Its Ir cannot be resolved: parent instructions retired of identical sessions drift by about 1 % (2.839 x 10^9 / 2.809 x 10^9 for two rho-only sessions) against a lab arm total below that, so the lab arm is bounded above by the noise floor, 3 x 10^7 per 8 targets = 3.75 x 10^6 per target.

4. Root extraction (instrument built for this pilot)

msolve 0.9.5 at characteristic 2: plain `solve` runs its FGLM step but writes an **empty** output file, and `-P 1|2` writes nothing either (checked on a toy system). Roots are therefore read from the `-g 2` reduced basis: the standard monomials give the quotient algebra A (dimension = number of F_2 points, the ideal is radical because the field equations are in it), the multiplication matrices of the x-bits are built by normal forms and A is split into joint eigenspaces (an FGLM-type step, `msolve_s4.rs`). Its native work is counted (`extraction_native_ops`, at most 4.8 x 10^3 per target, negligible against msolve) and **uncharged** in S. Checked against brute force over V^3 (every ordered triple with `S_4 = 0`): **14/14** targets at n = 9 in each form (6 non-unit ideals, largest 46 points), **3/3** at n = 13 (`probes/`). The unit-ideal verdicts (no relation) agree with brute force too.

5. Cells

`S` = GAE per sqrt(r), one target, cold. Lab S is the committed or session `ecbench` value, **solver uncharged**; the section 5a band prices the lab's word XORs at 0, 0.1, 1 GAE per word XOR.

cell	published-arm S	lab-arm S	ratio	verified	status
K_0 n=9, r=127, m=3, l=4, 8 targets	1.52e8 (1.05e8 to 2.27e8)	36.1 uncharged	4.2e6 uncharged; 2.7e4 at 0.1; 2.8e3 at 1 GAE/XOR	8/8	measured
(same cell) vs matched rho	2.6e7 times rho	rho 5.90; lab 6.1 times rho			
K_0 n=13, r=2003, m=3, l=5	not run; projection 7.6e9 (e_i), 1.7e9 (x-only)	16.29 uncharged	about 1e8 to 5e8 (projection)	one system checked	did not complete
other binary, Koblitz 41-61, m=4, prime n=4,5	not run				not run

Notes to the table.

- **n = 9 row.** Instance `icv1-f2m9-t5-81e744be` (#E = 508, cofactor 4), session `ECBS1h3c8ebd974409`. Published-arm S = (group GAE + Ir / 604.6) / sqrt(127), per target; median and range over the 8 targets. Lab S is the session's value (solver uncharged); with the lab's word XORs priced at 0.1 and 1 GAE per XOR (AGENTS.md section 5a) it is 5.5e3 and 5.5e4, so the order (msolve above lab) holds across the band. Against the instruction-level ceiling for the lab arm (3.75e6 Ir per target) the ratio is at least 2.8e5. All three arms verified 8/8 `[d]G = Q`; the msolve arm repeated gave Ir within 0.08 % and identical S.
- **n = 13 row.** One system took 5.70e12 Ir (544 s wall, loaded host); the x-only form 1.27e12 Ir (two systems). Projection = 36 systems per target (median solver calls of the committed K13 inherited-F4 records) x Ir per system / 604.6 / sqrt(2003). Lab 16.29 is the committed `SPEC-K13` value (uncharged); priced at 0.1 and 1 GAE per word XOR (median 1.1e7 XORs per target) it is 2.5e4 and 2.5e5, so the projected ratio is 7e3 to 3e4 against the band. An `ecbench` attempt at this cell was stopped at 5 min CPU and 4.5 GB RSS with the first system unfinished (`runs/C1_k13_aborted_process_time.txt`).
- **Not run.** Random binary: the symmetrised `S_4` is specialised to b = 1 (D6). Other sizes: the first system is already 1e12 to 1e13 Ir at n = 13.

Tuning probe (not a cell, per relation system, K_0): x-only form against `e_i` form, n = 9, l = 4 (14 targets): mean 1.21 x 10^10 against 6.49 x 10^10 Ir (5.4x); n = 13, l = 5: 1.27 x 10^12 against 5.70 x 10^12 (4.5x). Per-system lab cost: 4.0 x 10^4 word XORs (n = 9), 3.1 x 10^5 (n = 13). The cheaper form is still at least 3 x 10^5 times the lab at n = 9 if one word XOR is one instruction (it is more, so the ratio is lower; not resolved).

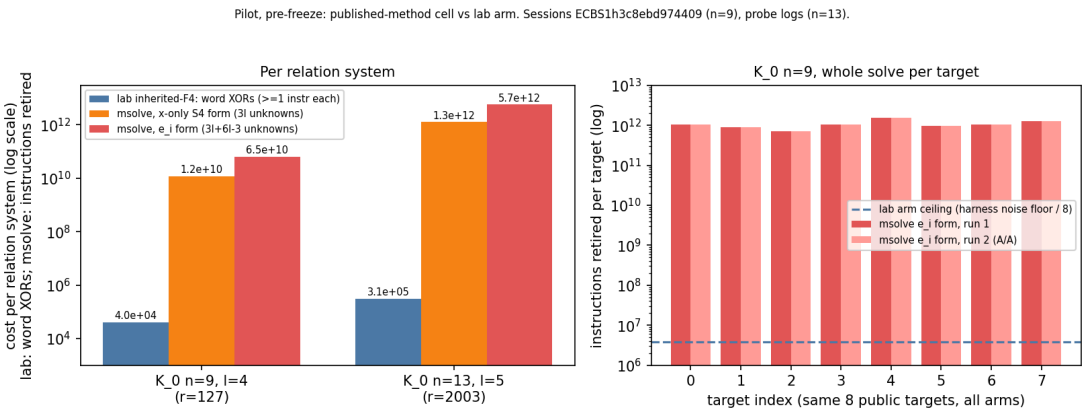


Figure 1: Per relation system and per target (recorded values).

6. Instrument defects found

- **D1. Worktrees share inodes.** `src/cryptanalysis/inherited_f4.rs` has 6 hard links across worktrees and `src/bin/ecbench/pdp3_koblitz.rs` 13. An in-place edit in one worktree appears in the others: this worktree showed another session's uncommitted `inherited_f4.rs` change (adds `mod profile;` with no `profile.rs`, so the tree would not build). *Evidence:* `stat -f %l`; `git diff` *Effect:* Edits here would have silently altered 12 other worktrees. I replaced my `pdp3_koblitz.rs` by a new inode before editing and **left `inherited_f4.rs` untouched and uncommitted** (not my change).
- **D2.** A sparse worktree cannot build: `Cargo.toml` names 40+ `[[bin]]/[[example]]` paths outside the sparse set, and `include_str!` reads `docs/` and `research/` files. *Evidence:* `cargo metadata` failures *Effect:* Built from a `git archive` subset (`src examples benches docs tests` plus 42 named files, plus `tools/volcano-cli`, `isogeny_algos`) in a separate directory; cold build 5.4 min.
- **D3. `ecbench` S is blind to an unrecognised solver unit**: the msolve arm and the lab arm carry the **same** S (35.494 on target 1), and the framework's own pre-removal figure treated msolve's wall nanoseconds as GAE (`framework_total_gae_before_solver_removal` 1.03 x 10^11). *Evidence:* `runs/D9` record, `cost.unpriced`, `detail` *Effect:* With a lower-bound S, an arm that is 10^6 times costlier looks equal. At n = 9 the lab's own solver is not small either: 6.2 x 10^5 word XORs per target against 400 group GAE.
- **D4.** `ecbench` holds one global lock `/tmp/crypto-bench.lock`; a second session exits at once unless `--wait` is passed. `isolation_required` must be `L0` here (`"none"` is rejected; `L1` cannot be earned). *Evidence:* run logs *Effect:* Specs in `specs/` use `L0`.
- **D5.** msolve 0.9.5 returns no solutions at p = 2 (section 4). `instructions retired` needs the sandbox off; the parent's figure excludes the msolve children; identical `ecbench` sessions differ by about 1 % in instructions retired. *Evidence:* `runs/A9`, `A9b` *Effect:* Per-call Ir comes from each msolve call, the lab arm is resolved only to 3 x 10^7 per 8 targets.
- **D6. Protocol defects.** (a) PROTOCOL section 1 says the reference is `e_i`-symmetrised, section 2 says "T-symmetrised"; the code has no `T` invariants. (b) B3/B4 list random binary curves, but `weil_descend_s4` asserts `b = 1`, so no random binary cell is runnable. (c) Section 4′s Callgrind control cannot run on this host. *Evidence:* `binary_semaev_s4.rs` *Effect:* Random binary Table B ids (15-27) cannot be reached by this arm; needs M1.

- **D7.** The conductor control plane at 127.0.0.1:8491 refused connections; no scope claim was made (hooks let edits through). `gh` and AWS credentials were not used; the checkpoint token worked. *Evidence:* tool errors *Effect:* None on results.

7. What is needed to go further

- **WDSat**: `github.com/mtrimoska/WDSat` at `61c6ff3f49445af2729274819edb27b85a89efc1` (GPL-3.0, external process, build from `src/` with `make`, per-cell `config.h` constants); not fetched, size not checked, absent from this host. Needs an instance emitter (M3).
- **A Linux host with Valgrind** for the protocol's Callgrind `Ir` and the cross-unit control.
- **msolve 0.6.5 built from source** (`research/dreg_closure_m4l6_20261018/pod/boot.sh`) for the version A/A against 0.9.5.
- **M1**: general-b `S_4` (random binary), the `T`-invariants, and `m = 4`; **tuning sweeps** (specialisation/hybrid, `l`, `LP`) so that the reference gets the tuning parity of PROTOCOL section 6. Without them the n = 13+ cells are not affordable (about 2 x 10^14 Ir per target projected for the e_i form).

8. Files

All under `pilot_20261030/`: `specs/` (the specs run), `runs/` (records, session, host, process time of every session, the aborted K13 attempt's process time), `probes/` (extraction check, n = 13 systems), `graphs/` (SVG, PDF, 150 dpi PNG, `numbers.json`, `fig_per_system_data.json`). Code: `src/bin/ecbench/msolve_s4.rs`, `examples/msolve_s4_probe.rs`, and the `msolve-s4` engine in `src/bin/ecbench/pdp3_koblitz.rs`. Derived values in section 5 are arithmetic on the recorded values, shown in the table; they were computed with a throwaway script, not committed as research code.